

Bonus 1: A scientific Ohm-law figure

Physics Programming, Week 1 — Optional extra credit

Optional bonus: make a scientific figure communicate

This bonus is optional, is not a prerequisite for later work, and is not examined. A completed bonus is worth **one-third of a final-grade point**. Before you start, read this week's lecture together with the self-contained reading below; the optional *Going further: Lissajous curves* companion is not needed for this bonus.

After 10–15 minutes of self-contained reading, we improve a supplied minimal Ohm-law figure and compute its numerical fit. Here voltage is the horizontal coordinate and current the vertical coordinate, so the model slope is conductance in S, not resistance in Ω . We identify the measured-current series and the fitted-current series before changing their appearance.

We add the prescribed title, labels with units, legend entries, observation markers, model line, frame, and canvas size, then export PNG and PDF. The graded fragment is numerical only: it returns the fitted conductance and the aligned fitted and residual current series. The figure itself is a separate artifact assessed against the explicit checklist, because a graded kernel does not assume plotting support. Changing a label or line style must not alter a measurement.

Scope notice

This bonus is optional, is not a prerequisite for later work, is not examined, and is worth **1/3 point on the final course grade**.

Self-contained reading (10–15 minutes)

Physics box: an ohmic conductor

Voltage drives a current through a conductor. In the ohmic regime the current is proportional to the applied voltage, so $V = RI$, equivalently $I = gV$. Voltage V is in volts, current I in amperes, resistance R in ohms (Ω), and conductance $g = 1/R$ in siemens (S). We take the positive current in a chosen reference direction through the conductor; reversing that direction flips the sign of both I and the fitted intercept. The relation assumes a fixed conductor state, a weak field, and linear response, so R stays constant over the measurement. The ideal check is $V = 0 \Rightarrow I = 0$. Heating changes R , strong fields and non-ohmic devices bend the curve, and contacts create offsets, so a straight line fitted here holds only over the measured range. Scattering establishes steady drift and helps set the resistance, but rapid collisions alone do not guarantee linearity.

With current on the vertical axis and voltage horizontal, a fitted line $I = gV + I_0$ has slope g equal to conductance in S and intercept I_0 in A. The ideal check is $V = 0 \Rightarrow I = 0$.

Task: the graded numerical fragment

The platform supplies aligned vectors `voltage_V` and `current_A`. Using the same centered-sum recipe as the Week 1 lecture, fit $I = gV + I_0$ and leave three names: `conductance` (the slope in S), `fitted` (the fitted current at every supplied voltage), and `residual` (measured minus fitted current). Submit statements only: no `function`, `return`, `validation`, `printing`, `file reading`, or `plotting`.

```

V_bar = sum(voltage_V) / length(voltage_V) # V
I_bar = sum(current_A) / length(current_A) # A
dV = voltage_V .- V_bar # V
dI = current_A .- I_bar # A
conductance = sum(dV .* dI) / sum(dV .^ 2) # S
intercept = I_bar - conductance * V_bar # A
fitted = conductance .* voltage_V .+ intercept # A
residual = current_A .- fitted # A

```

Task: the figure artifact (assessed separately)

Write a short plotting program that produces one figure with the exact title **Ohm-law measurements and fitted model**; horizontal label **Voltage (V)**; vertical label **Current (A)**; observation-marker legend **measurements**; model-line legend **least-squares model**; boxed frame; and canvas size (800,500) pixels. Save it as both PNG and PDF, and include the numerical model and one residual or known-case interpretation in your short report. Both series share the same vertical axis, current in A, because the fragment returns the fitted current at each supplied voltage.

```

p = scatter(voltage_V, current_A; label="measurements",
    xlabel="Voltage (V)", ylabel="Current (A)",
    title="Ohm-law measurements and fitted model", framestyle=:box, size
    =(800, 500))
plot!(p, voltage_V, fitted; label="least-squares model")
savefig(p, "ohm_law.png")
savefig(p, "ohm_law.pdf")

```

How is the fragment checked?

To check the numeric fragment, the public tests feed it a supplied fixture and compare `conductance`, `fitted`, and `residual` with `isapprox` ($\text{atol} = \text{rtol} = 10^{-10}$). The grader receives the data directly, so it makes no assumption about a local file, and plotting plays no part in this half of the grade. Its own validation turns away data with mismatched lengths, non-finite or Boolean values, or no spread in voltage. The staff tests rest on the same published contract, so what we see here is what our fragment is graded against.